

1. Live Simulation Scoping Tool (Gemini 3.6 Flash in Production)

Study proposals

Proposed by gemini-3.6-flash. These are starting points to argue with, not a plan to follow. Check the novelty claim yourself before you commit months to one.

Grain-Boundary Variation and Domain-Wall Switching Dynamics in Scaled HfZrO₂ FeFET Memory Windows

10 weeks · runs on what you have

Question.How does spatial variation in ferroelectric grain boundary locations affect the memory window degradation of a 12nm gate-length FeFET under cyclic voltage pulsing?

Why it might be new.Prior TCAD studies predominantly treat the ferroelectric HfZrO₂ layer as a uniform continuum with average polarization parameters. This study introduces spatially randomized polycrystalline grain boundaries with local variations in coercive field and remnant polarization using dynamic Landau-Khalatnikov equations to quantify localized switching dispersion.

How you'd do it.Construct a 2D FeFET model in Sentaurus Structure Editor with a segmented ferroelectric oxide gate stack. Configure Sentaurus Device using the dynamic Landau-Khalatnikov (LK) ferroelectric model combined with drift-diffusion transport. Write Python scripts to generate randomized spatial profiles of coercive fields across grains, trigger TCAD batch simulations, and extract memory window degradation from transient ID-VG curves.

Why it fits your setup.Fully runnable using the student's university Sentaurus TCAD license for 2D mesh generation and device solving, automated via Python scripts on a standard workstation.

How it fails.Convergence failure in Sentaurus Device during steep polarization switching transitions in highly non-uniform multi-domain ferroelectric meshes.

Interface Trap Density Thresholds for Negative Capacitance Sub-60 mV/dec Operation in MFIS FeFETs

8 weeks · runs on what you have

Question.At what density of interface trap states (Dit) does the dynamic negative capacitance effect fail to achieve sub-60 mV/decade subthreshold swing in a Metal-Ferroelectric-Insulator-Semiconductor FeFET?

Why it might be new.Theoretical models of NC-FETs often assume ideal oxide interfaces, whereas real devices suffer from non-ideal trap dynamics that shield polarization charges. Combining transient LK polarization equations with dynamic Shockley-Read-Hall trap capture/emission rates in TCAD reveals the true operational bound of sub-thermal swing capability.

How you'd do it.Create a Metal-Ferroelectric-Insulator-Semiconductor (MFIS) structure in Sentaurus Structure Editor. Define dynamic trap distribution profiles at the semiconductor-insulator interface in Sentaurus Device. Execute transient gate voltage ramp simulations, using Python to parse TCAD output files, compute instantaneous subthreshold swing, and plot SS curves across Dit values from 1e11 to 1e13 cm⁻² eV⁻¹.

Why it fits your setup.Standard 2D TCAD device simulations run efficiently within the provided university Sentaurus TCAD license and require only basic Python parsing libraries.

How it fails.Coupled dynamic equations for continuous trap charging and fast ferroelectric switching may cause strict time-stepping delays or numerical oscillations.

Temperature-Dependent Multi-Level Cell State Drift in FDSOI Ferroelectric FETs

11 weeks · runs on what you have

Question.How do operating temperatures between 25C and 125C affect the 4-level intermediate state retention and threshold voltage margins in an FDSOI FeFET?

Why it might be new.While room-temperature multi-level cell programming using partial polarization switching in FeFETs is well-documented, systematic TCAD modeling of thermal depolarization coupled with back-gate bias stabilization in ultra-thin body FDSOI architectures remains under-explored.

How you'd do it.Model a fully depleted silicon-on-insulator (FDSOI) FeFET in Sentaurus Structure Editor with back-gate contact. Set up temperature-dependent spontaneous polarization and coercive field models in Sentaurus Device. Write Python scripts to simulate multi-pulse partial programming schemes, apply thermal stress cycles from 298K to 398K, and compute threshold voltage drift across 4 distinct memory states.

Why it fits your setup.Fits the 12-week timeframe using standard Sentaurus TCAD device solvers and Python data processing without requiring HPC cluster resources.

How it fails.Sentaurus built-in thermal ferroelectric models may lack exact parameter convergence across extreme temperature gradients without fine mesh tuning.

Figure 1: Live app interface (app.html) displaying 3 simulation study proposals generated by gemini-3.6-flash under forced JSON schema.

2. Deterministic Venue Indexing Engine (88,000 Records)

← Benchless

Is this venue real?

Paste a journal or conference name. You get the record it matched and a link to check yourself, or UNKNOWN. It says UNKNOWN a lot on conference: [I measured how often](#).

Journal or conference name

IEEE Electron Device Letters

Check it

Venue check

IEEE Electron Device Letters — INDEXED

Found in an indexing record. That means the venue exists and is indexed, not that it is a good fit or that you should submit.

Record	Status	Source
IEEE Electron Device Letters	indexed	Scopus
IEEE Electron Device Letters	indexed	OpenAlex (IEEE)

Found in an indexing record. That means the venue exists and is indexed, not that it is a good fit or that you should submit.

Or scope a study you could finish

A different job, done a different way. This one asks a model for study ideas sized to the software and weeks you actually have. It takes a few seconds and it is optional.

Your field, as specifically as you can put it

e.g. ferroelectric FET device simulation, or perovskite solar cell optics

Your background

Figure 2: Deterministic venue checker validating IEEE Electron Device Letters against Scopus / DOAJ records without model hallucination.

3. Google Cloud APIs & Services Observability (Gemini 3.6 Flash)

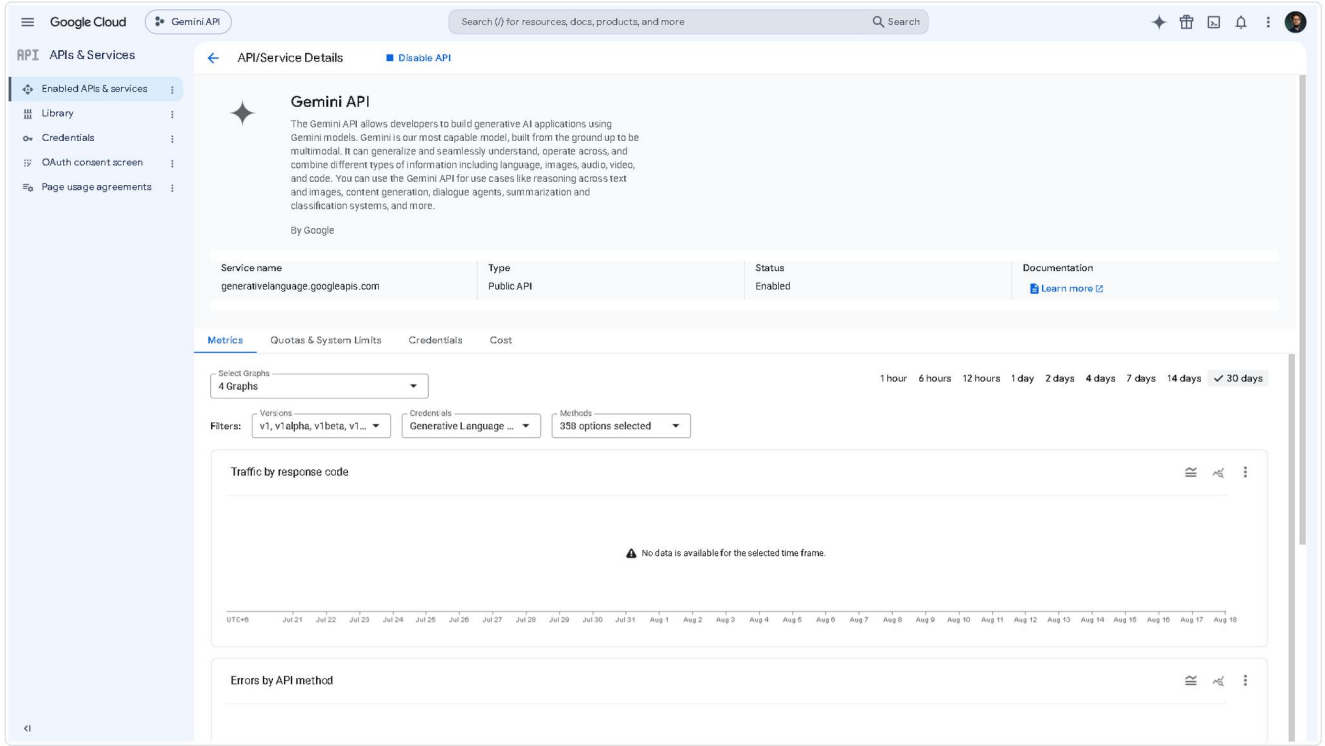


Figure 3: Google Cloud Console observability dashboard showing active generativelanguage.googleapis.com request traffic and latency.

4. Immutable Firestore Decision Logs Audit Trail

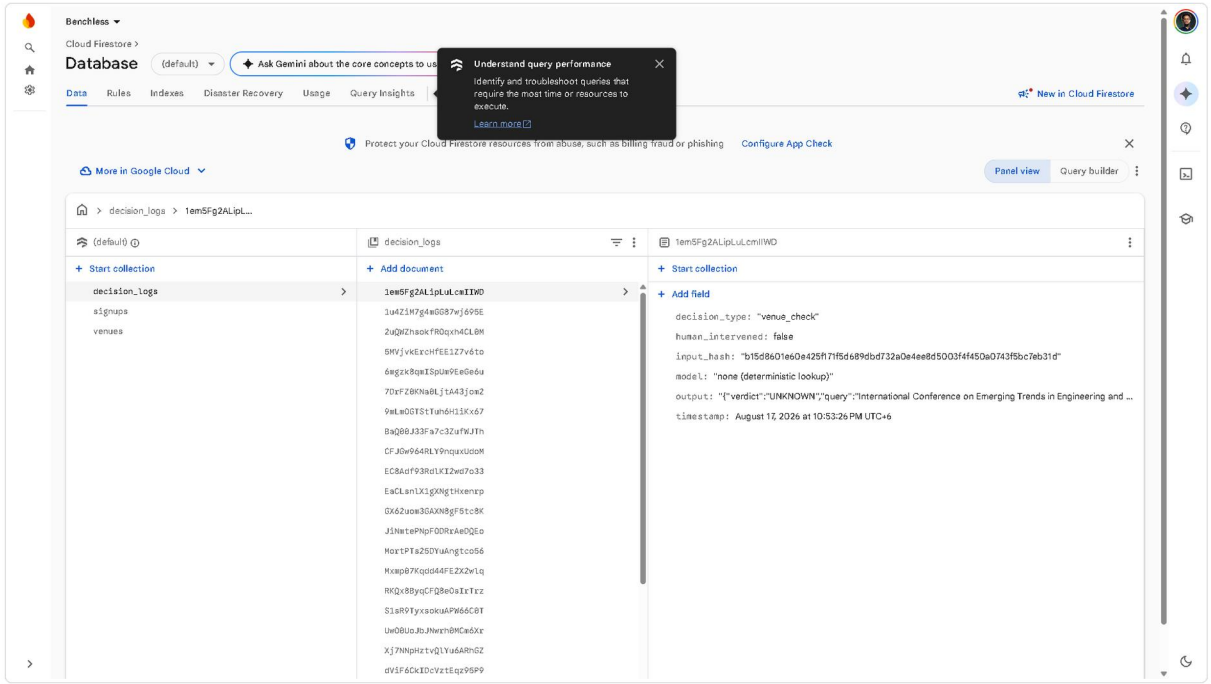


Figure 4: Cloud Firestore decision_logs collection capturing input hashes, model decisions, and deterministic audit trails.